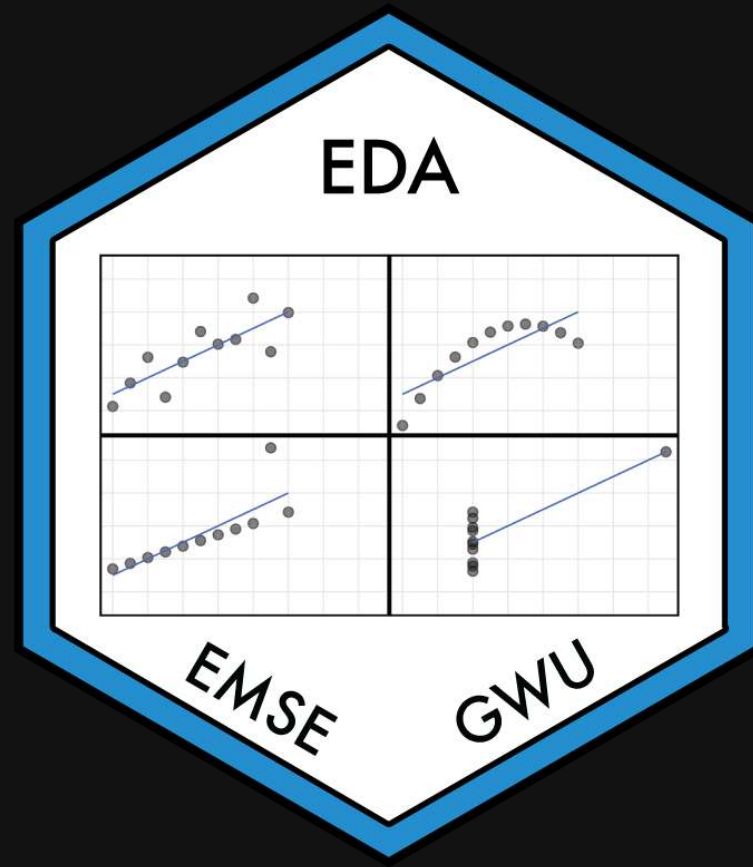




Agentic Workflows & GitHub

🏛️ EMSE 4572/6572: Exploratory Data Analysis

👤 John Paul Helveston

📅 September 2, 2026

Week 2: *Agentic Workflows & GitHub*

1. Version Control with GitHub
2. Setting Up & Using Claude Code
3. Modes & Slash Commands
4. Skills
5. Working Safely

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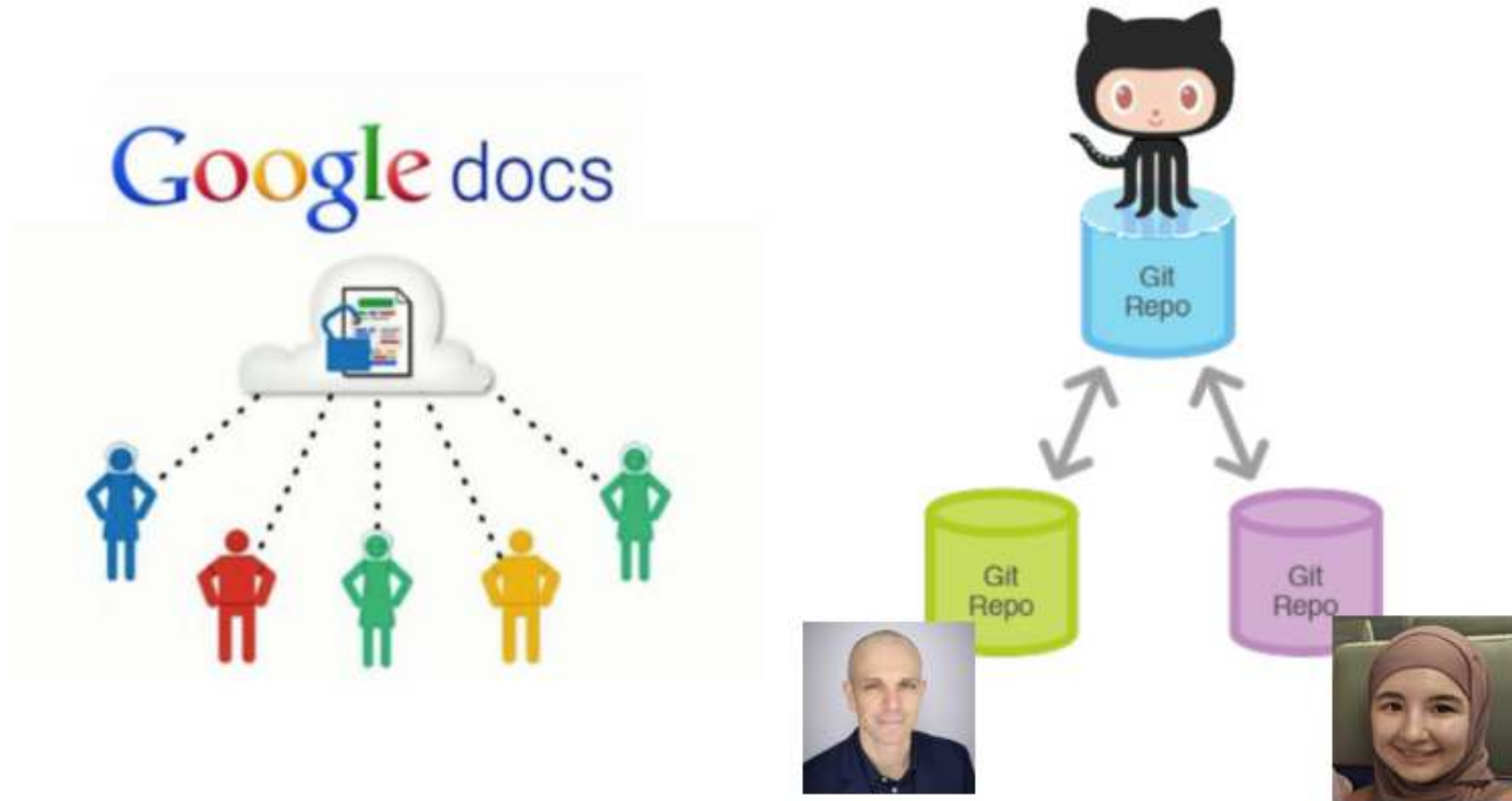
VCS = Version Control System

Original domain: **software development**

Git = The most popular VCS

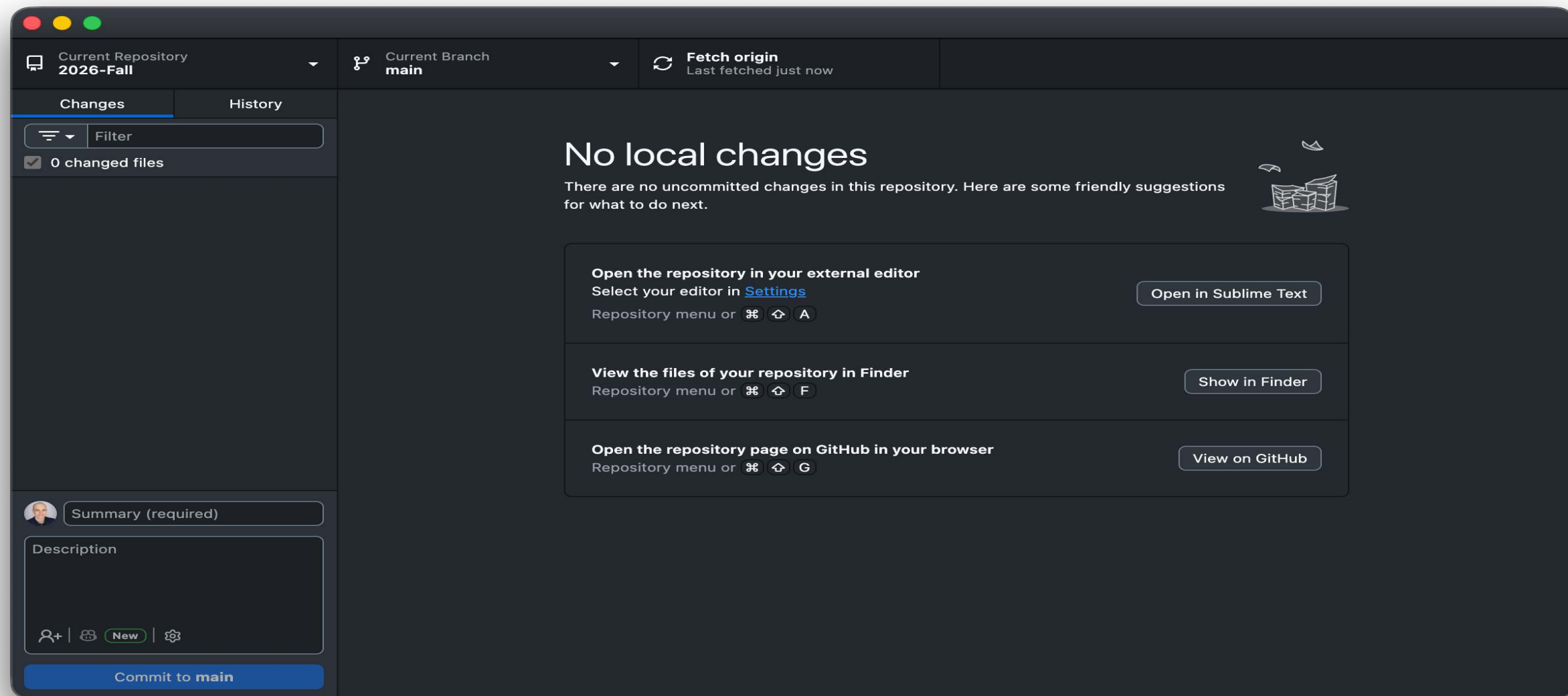
Our domain: **data analysis & agentic workflows**

We collaborate using git on github.com

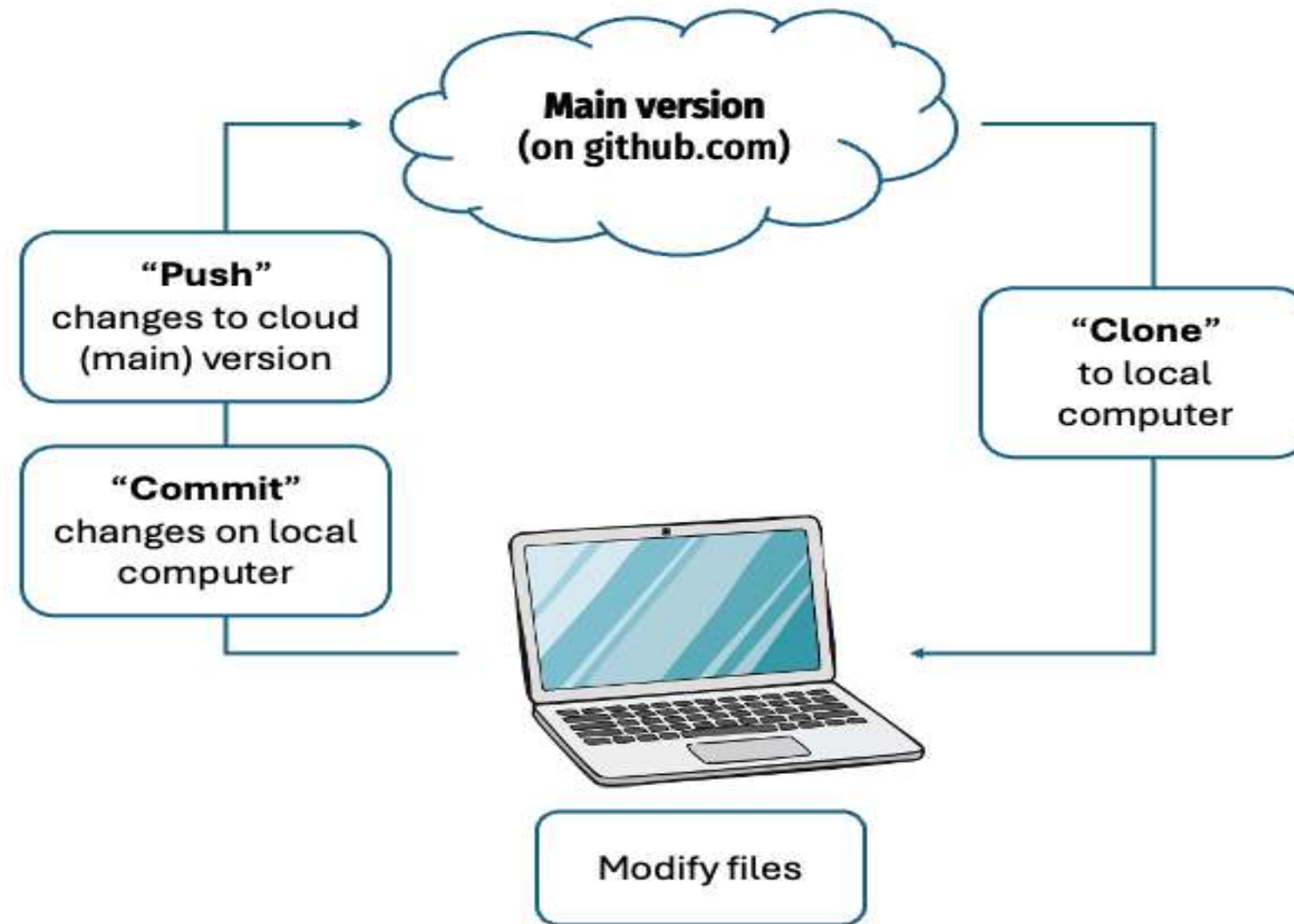


We won't use `git` in a terminal
(you can if you want to)

Everything is a **button** in GitHub Desktop



The github workflow



Git vocabulary you should know

“repo”

A project folder that **tracks its own history**

Example: course website:
<https://github.com/emse-eda-gwu/2026-Fall>

“commit”

A **labeled save point** —
“this is a state worth remembering”

“push” / “pull”

Send / receive your commits **up to GitHub**
(the cloud copy)

The GitHub Desktop loop

Changes (locally on your computer) → **Commit** (with a message) → **Push**

Write commit messages you'd thank yourself for

✗ Useless

- `stuff`
- `update`
- `asdf`
- `fixed it`

✓ Useful

- `Add flights bar chart`
- `Fix airline join dropping NAs`
- `Clean column names in import`

A message is a note to **future you**

Checkpoint 👍

You already ran the solo loop in **HW1** — let's make sure it stuck.

1. In **GitHub Desktop**, confirm your **eda-practice** repo is there
2. Open it on **GitHub.com** — can you see your commit in **History**?
3. **Thumbs up** once you can see your pushed change online

Missing it, or joined the class late?

Flag me now — grab the steps from [HW1](#) or this [2-min walkthrough](#)

How do I collaborate with others?

Add collaborators in repo settings

The screenshot shows the GitHub repository settings page for 'surveydown-dev/surveydown'. The 'Settings' tab is selected in the top navigation bar. In the left sidebar, the 'Access' section is expanded, and 'Collaborators and teams' is highlighted. The main content area is titled 'Collaborators and teams' and shows the repository is public. It includes sections for 'Base role' (Write), 'Direct access' (3 members), and 'Organization access' (0 users, 0 teams). Below these is the 'Manage access' section, which is highlighted with a red box. This section has tabs for 'Direct access' and 'Organization access'. Under 'Direct access', there is a 'Select all' checkbox and a search bar. A list of collaborators is shown, including Bogdan Bunea, John Helveston, and Pingfan Hu, each with a role of 'write' and a trash icon for removal.

Manage access

Create team Add people Add teams

Direct access Organization access

Select all Type Role Find people or a team...

☐	Bogdan Bunea Buneabt		Role: write	
☐	John Helveston jhelvy	⚠ Mixed roles	Role: write	
☐	Pingfan Hu pingfan-hu	⚠ Mixed roles	Role: write	

Two new ideas

“clone”

Make your own **local copy**
of someone else's repo

“pull request” (PR)

Propose your changes back
they review, then **merge**

You work on a **branch**, so their **main** is untouched
until they say *yes*.

Why work on a branch?

On a branch

- Their `main` stays **untouched**
- You experiment freely
- Nothing lands until it's **approved**

Straight to `main`

- Your half-done work **is** the project
- No review step
- Mistakes hit everyone at once

A branch is a **sandbox**. The pull request is where they say *yes, bring it in*.

New Branch → **Commit** on branch → **Make PR** → **Merge** to main

Pull request loop

Your turn

10:00

Team up and send your partner a pull request

1. Pair with a neighbor. In *your* **eda-practice** repo on GitHub.com:
Settings > **Collaborators** → add your partner's GH username → they **accept** the emailed invite
2. Swap repo links. In GitHub Desktop, **File** > **Clone Repository** → clone *your partner's* repo
3. **Branch** > **New Branch** → name it **new-edits**
4. Edit their **README.md** in Positron → **Commit** to your branch → **Publish branch**
5. Click **Create Pull Request** → describe your change → submit it on GitHub.com
6. Back on *your* repo, review your partner's PR and **Merge** it

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Assistants vs. Agents

This class lives on the **agent** side.

Agentic Workflow

Claude Code is the agent. **You** supply the project and the direction.

An agent can do almost anything with files



Write code



Debug & refactor



Analyze data



Generate images



Summarize docs



Write & edit prose



Automate tasks



Build HTML

Claude Code is an agent (a particular “harness”)

Other agents:

Gemini CLI

Google's open-source terminal agent

Codex

OpenAI's coding agent (CLI + cloud)

Using Claude code in Positron

1. Make a folder (no spaces in the name)
2. In Positron: **File › Open Folder...** and pick it
3. In the terminal pane, type `claude` and press Enter

Your first agentic project: a Quarto website

10:00

Direct the agent to build a multi-page Quarto website about **someone you admire (a scientist, athlete, artist, etc.)**

1. Using GitHub Desktop, make a new repo called `my-website` like you did in the HW (include a `README` file). Make it `public` (not `private`), and publish it to GitHub.
2. On your computer, open Positron, then open the folder of the repo.
3. Launch `claude` in terminal.
4. Prompt Claude for a multi-page **Quarto website** about someone you admire.
5. When it's done, ask it to render the site (if it didn't already).
6. Open the `index.html` file in the `_site` folder.
7. If satisfied with the site, **commit + push** it to GitHub.

Extra: publish your site with GitHub Pages

1. Ask Claude to change the Quarto output folder from `_site` to `docs` (edit `_quarto.yml`'s `output-dir`), then re-render.
2. **Commit + push** your changes to GitHub.
3. On GitHub.com, go to your repo's **Settings** > **Pages**.
4. Under **Build and deployment**, set **Source: Deploy from a branch**, **Branch: `main`**, **Folder: `/docs`** → **Save**.
5. Wait a minute, then visit the URL GitHub shows you at the top of the Pages settings page.

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Four modes of Claude Code



Make a plan before doing big work.

read-only



Approve or deny each edit.

safe but slow



Auto-applies file edits.

semi engagement



Approves EVERYTHING.

full file access, use with care

Press **shift + tab** to cycle through them

The current mode always shows at the **bottom of the prompt box**

Modes are your permission dial

plan → manual → accept edits → auto

less freedom, more supervision

more freedom, less supervision

Give more autonomy to Claude when
you have a *very* clear step-by-step plan

More control over session with slash commands

Five you'll reach for constantly:

`/init` · `/model` · `/effort` · `/clear` · `/compact`

CLAUDE.md: standing instructions

- A plain markdown file the agent **reads at the start of every session**
- It's your project's **house rules** — what this project is, how to run it, what you want done differently

Project CLAUDE.md

rules for *this* repo, shared with anyone who clones it

Home ~/.claude/CLAUDE.md

rules for *everything* you do, on your machine only

`/init` — let the agent write `CLAUDE.md` for you

You don't write `CLAUDE.md` from scratch.

When you run `/init`, Claude:

1. Walks the folder — file tree, `README`, config files, existing code
2. Works out **what the project is** and **how it gets built or run**
3. Writes `CLAUDE.md` in the project root

What `/init` actually produces

A short file, roughly:

```
1 # CLAUDE.md
2
3 ## Project overview
4 A multi-page Quarto website about Ada Lovelace.
5
6 ## Building
7 `quarto render` builds the site to `_site/`.
8 `quarto preview` for live preview.
9
10 ## Structure
11 - `index.qmd` – home page
12 - `_quarto.yml` – site config & navbar
```

`/init` is a *draft*, not gospel

Claude is **guessing** from what it can see. It gets things wrong.

⚠ Always read it

- Is the description right?
- Are the build commands real?
- Did it invent a convention you don't follow?

✎ Then edit it by hand

- Delete what's wrong or obvious
- Add rules it can't know:
 "charts use `theme_minimal_hgrid()`"
- Re-run `/init` only after big changes

/model — pick the right **brain** for the job

Model	Good for
Opus	Hard reasoning, tricky bugs, big refactors
Sonnet	Everyday work — fast, capable, cheaper
Haiku	Quick, simple, high-volume tasks

/model switches anytime. Bigger isn't always better.
Match the model to the task.

`/effort` — how *hard* the model thinks

Same model, different amount of thinking before it acts.

Level	What it's for
<code>low</code>	Simple, mechanical edits — fastest, cheapest
<code>medium</code>	Everyday work
<code>high</code>	Multi-step tasks, debugging, design decisions
<code>xhigh</code>	Genuinely hard problems — slowest, most expensive

`/effort auto` returns to your model's default.

/model vs. /effort

/model

Which brain.

Changes the underlying capability — reasoning ability, speed, cost per token.

/effort

How long it thinks.

Same brain, more deliberation before it starts editing files.

General rules of thumb:

- Turn effort **up** when the hard part is *deciding what to do*
- Turn effort **down** when **you already know exactly what you want**

Tokens are the currency

A **token** is a chunk of text — roughly $\frac{3}{4}$ of a word.
Everything the agent does is paid for in tokens.



Thinking

Reasoning before it acts



Running tools

Every file it reads, every
command it runs



Writing code

Everything it writes back to you

Your budget refills on a clock



Every 5 hours

The clock starts at your **first message**, then resets.

the one you'll actually hit



Every 7 days

A **weekly cap** sitting on top of the 5-hour one.

the safety net

Run out mid-assignment and you *wait*

Mind your context

Every message re-sends the **whole conversation**, not just your last line.



0–60%

Room to think

healthy



60–80%

Slower and pricier

trim it soon



80–100%

Hard stop at 100%

start fresh

Two consequences: a **tiny question pays for the whole history**,
and you should **start fresh early**, not at 99%.

Fewer words in, fewer words out

The agent's *chattiness* is billed too — so tell it to be terse.

The 🪨 **Caveman** plugin does exactly that:

“why use many token when few do trick”

Normal agent

“The reason your React component is re-rendering is likely because you’re creating a new object reference on each render cycle. When you pass an inline object as a prop, React’s shallow comparison sees it as a different object every time, which triggers a re-render. I’d recommend using `useMemo` to memoize the object.”

69 tokens

With Caveman

“New object ref each render. Inline object prop = new ref = re-render. Wrap in `useMemo`.”

19 tokens

/clear vs. /compact — managing memory

/clear

Wipe the slate.

- Forgets the whole conversation
- Fresh start, empty context
- Use it **between unrelated tasks**

/compact

Summarize and continue.

- Condenses the chat so far
- Keeps the gist, frees up room
- Use it **mid-task** when things get long

Your turn

10:00

Go back to your **my-website** repo

1. Run **/init** — then **open the CLAUDE.md it wrote and read it**. Is it right? Fix anything wrong, and add one rule of your own.
2. Run **/model** and **/effort** to see your options.
3. Ask for a **small** change (new page, tweak the navbar) at **/effort low**.
4. Hit **shift + tab** into **plan mode**, then ask for something harder, e.g., *"collapse everything into a single page website"* at **/effort high**. Read the plan before you approve it.
5. **/compact**, then keep going — it should still know what you're building.
6. **/clear**, then ask *"what am I working on?"* — see what it lost, and what **CLAUDE.md** saved.
7. **Commit + push** your changes to GitHub.

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Skills: a command that wraps a whole recipe

`/init`, `/clear`, `/effort`, etc. are **built in** with Claude Code

A **skill** is a slash command *you* define that teaches the agent one of **your** recipes.

Why they help

- 🕒 Save time
- ♻️ Reproducible
- 🛡️ Stable & consistent

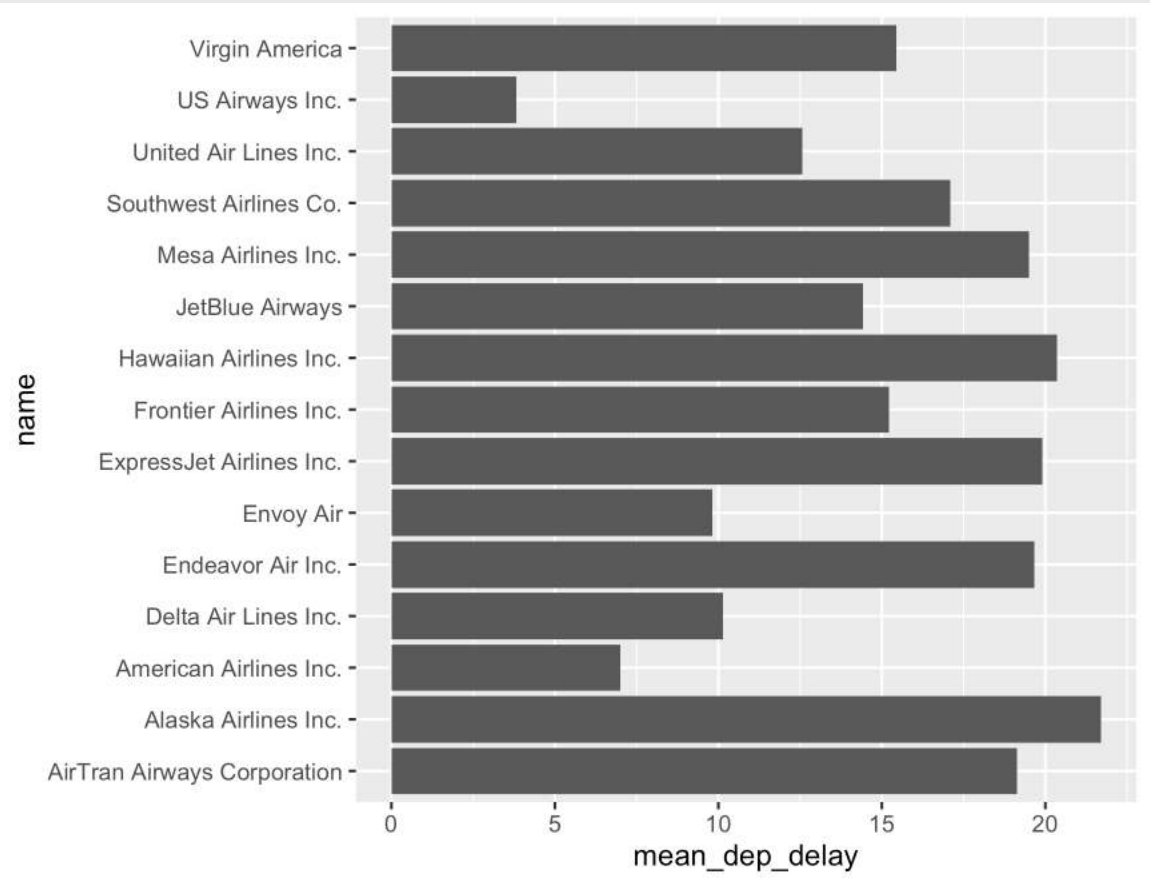
Using one

- Live in `~/.claude/skills/`
(or `.claude/skills/` in a repo)
- Invoke with `/skill-name`
- We'll try one for **charts**: `/my-chart-style`

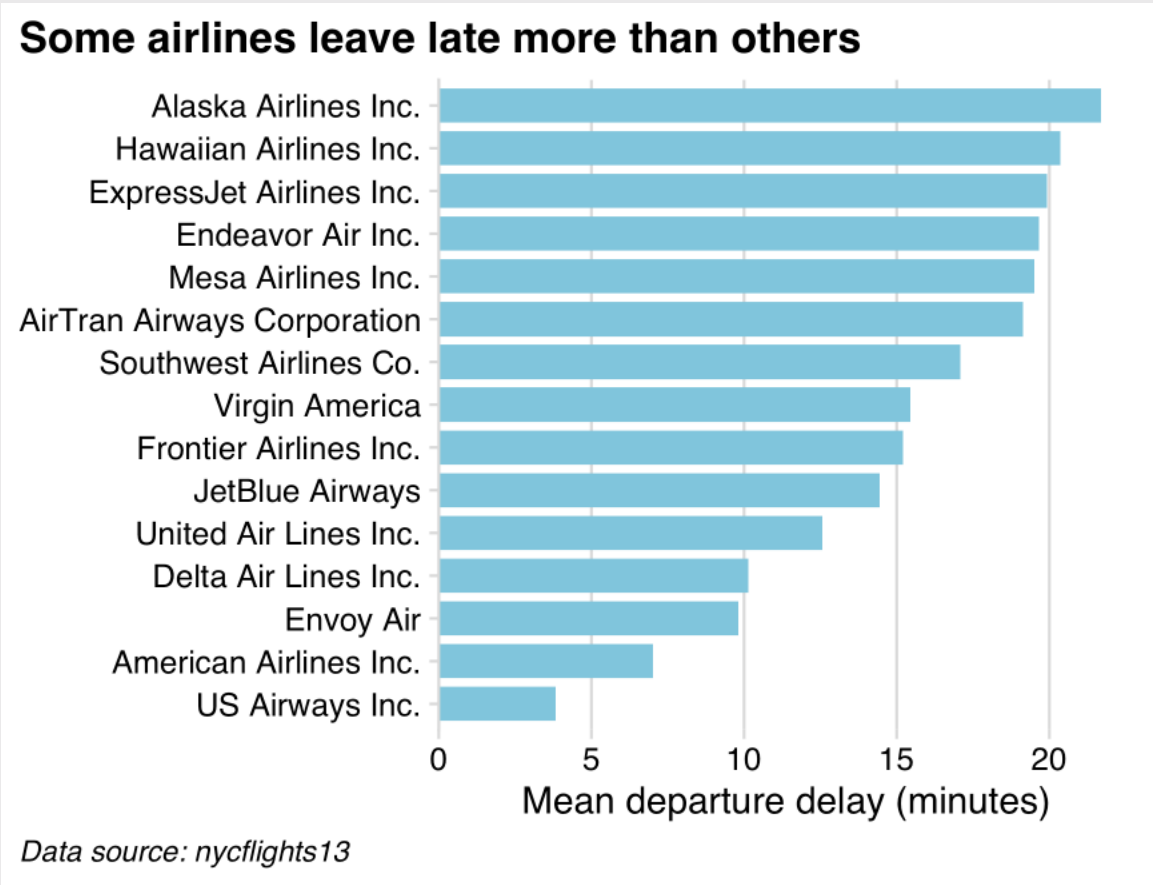
A skill in action: `/my-chart-style`

Same request — *“mean departure delay by airline, as a bar chart”*:

Without the skill



With the skill



What this skill defines

- a `cowplot` package minimal theme (gridlines matched to the plot)
- white background, de-cluttered gridlines
- a consistent color palette + sorted bars
- a title / caption that state the point and the source

Learn the conventions once → **encode them in a skill**

Your turn

10:00

See the difference a skill makes

1. Ask Claude to copy the `my-chart-style` skill from today's class folder into `.claude/skills/` in your project (create the folder if it doesn't exist)
2. Come up with a chart you'd like to see that uses the `flights.csv` data.
3. Ask Claude for a chart first **without** the skill, then again invoking `/my-chart-style`.
4. Modify the skill, then re-create the chart again with the modified skill (e.g., tell it a specific color to use for all charts, tell it to use a different theme, etc.)

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Agents will be *confidently wrong*

They won't say *"I'm not sure"*

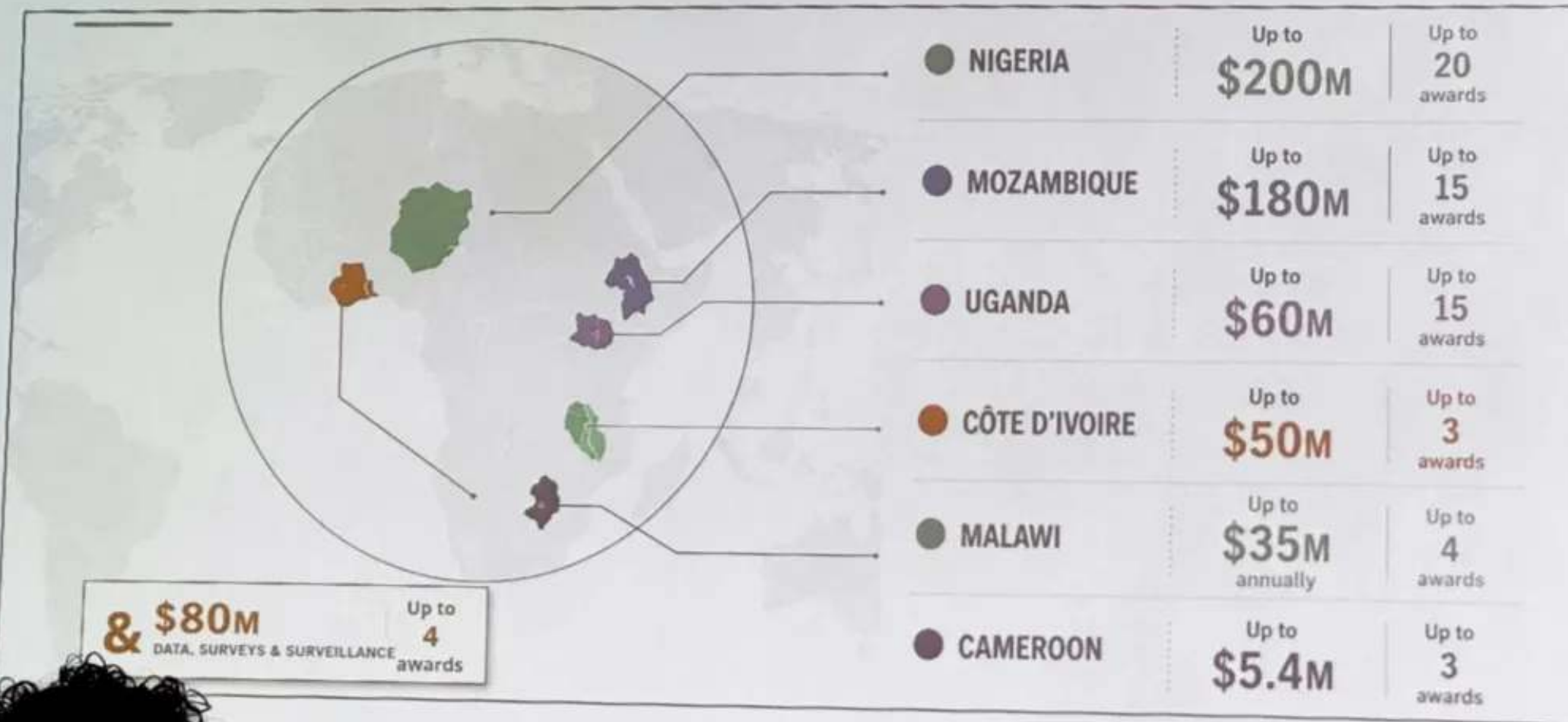
Fast $\neq$ Better

Throughput goes up, but **quality is not guaranteed**

Defense Strategies

1. Ask for code, not results
2. Run sanity-checks
3. Use fake data

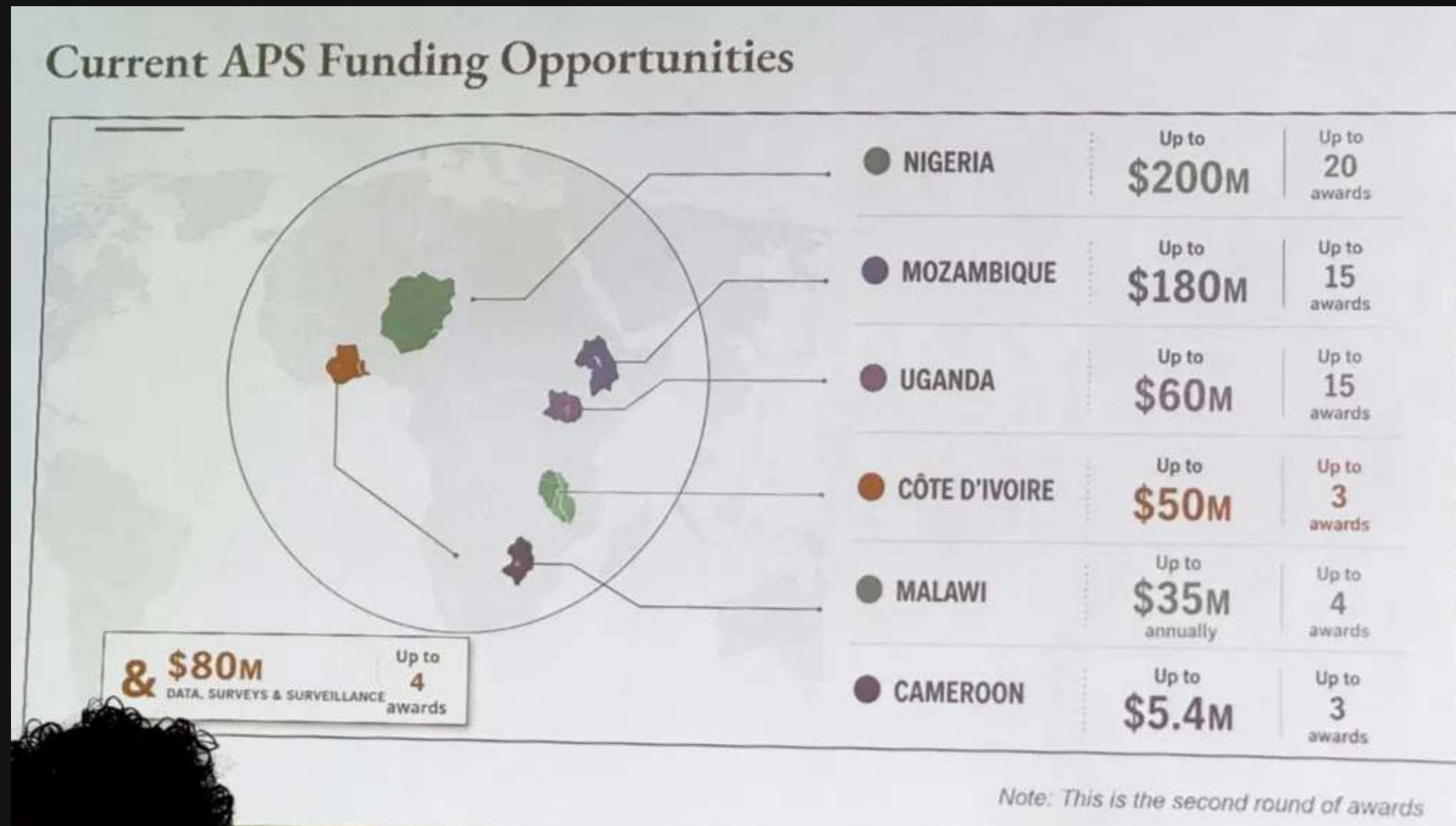
Current APS Funding Opportunities



Note: This is the second round of awards

Source: [BBC](#)

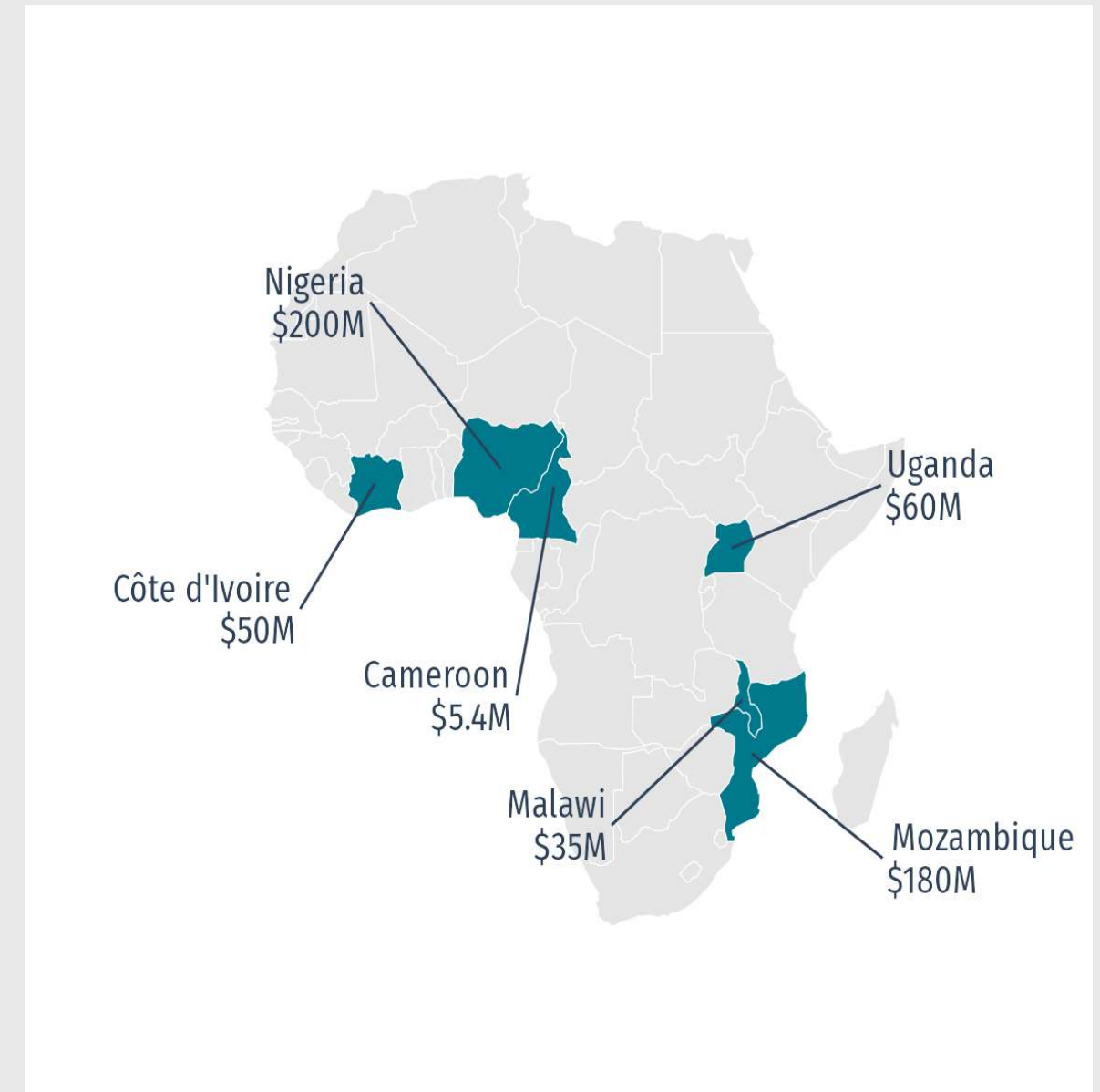
There is no version that throws an error



Source: [BBC](#)

Same map from source

```
1 library(tidyverse)
2 library(rnaturalearth)
3 library(sf)
4
5 funding <- read_csv("data/africa-funding.csv")
6
7 africa <- ne_countries(continent = "Africa", returnclass = "sf")
8 stopifnot(all(funding$name %in% africa$name))
9
10 funded <- africa |>
11   inner_join(funding, by = "name") |>
12   mutate(
13     lon = st_coordinates(st_point_on_surface(geometry))[, "X"],
14     lat = st_coordinates(st_point_on_surface(geometry))[, "Y"]
15   )
16
17 ggplot(africa) +
18   geom_sf(fill = "grey90", color = "white", linewidth = 0.2) +
19   geom_sf(data = funded, fill = "#00798c", color = "white") +
20   geom_segment(
21     data = funded, color = "#2e4057",
22     aes(lon, lat, xend = lon + dx, yend = lat + dy)
23   ) +
24   geom_text(
25     data = funded, color = "#2e4057", size = 5, lineheight = 0.9,
26     family = "Fira Sans Condensed",
27     aes(
28       lon + dx, lat + dy, hjust = ifelse(dx > 0, -0.05, 1.05),
29       label = paste0(name, "\n$", amount, "M")
30     )
31   ) +
32   coord_sf(xlim = c(-38, 66), ylim = c(-36, 38)) +
33   theme_void()
```



The pipeline hasn't changed

`data.csv` → `script.R` → `figure.png`

Same workflow, with or without AI

By hand

- Set up the project
- Build the scripts
- Run it
- **Verify the output**

With AI

- Set up the project
- Build the scripts
- Run it
- **Verify the output** ← Still yours

AI can do the first three. **Verifying the output is still yours.**

Agents rarely make a syntax mistake

- No typos
- No missing brackets
- A tidy `input/ scripts/ output/` layout

But **running** isn't the same as **right**.

You are still the gatekeeper for the results.

Defense Strategies

1. Ask for code, not results
2. Run sanity-checks
3. Use fake data

Verify against an *independent* check

Don't ask the agent if it's right — it'll say yes.
Check it a way it **can't fake**:

 A published number

Codebook, prior paper,
official table

 A hand calculation

Back-of-envelope,
done by *you*

 A second analyst

Different chat, different
prompt, different model

Your verification checklist

Run these every time — they're the skill this whole course is really about:

- Did **row / column counts** change the way I expected?
- Did I **read the code**, not just the summary?
- Did I **spot-check** a few values against the source file?
- Were rows **silently dropped** (NA handling, inner vs. left join)?
- Do **totals / aggregates** make sense?
- Is it **reproducible**, or did it hard-code an answer?

What a sanity-check script looks like

- ✓ row count survived the merge — 1,200 → 1,200
- ✓ every ID appears once — 0 duplicates
- ⚠ a percentage above 100% — check the units

Two checks passed silently. **One caught something real.**

Defense Strategies

1. Ask for code, not results
2. Run sanity-checks
3. Use fake data

Sometimes you *can't* hand over the data

- Medical records.
- Student grades.
- Proprietary sales figures.
- Anything under an NDA or an IRB protocol.

Make **fake data with the same shape**, and work on that instead.

The fake-data workflow

1. Build a **small fake table** with the same columns and types as the real thing
2. Have the agent write the script against **that**
3. Run the *same script* on the real data — on your machine

The agent never sees the real data. **The script still works on it.**

charlatan: fake data in R

```
1 install.packages("charlatan")
2 library(charlatan)
3
4 ch_name()
5 #> "Dr. Garey Hamill"
6
7 ch_job()
8 #> "Corporate investment banker"
9
10 # reproducible, like any RNG in R
11 set.seed(42)
```

```
1 # a whole data frame at once
2 ch_generate(
3   "name",
4   "job",
5   "phone_number",
6   n = 5
7 )
8
9 #> # A tibble: 5 x 3
10 #>   name                job                phone_number
11 #>   <chr>              <chr>              <chr>
12 #> 1 Dr. Garey Hamill Corporate inves... 1-5
13 #> 2 Elmer Rippin      Herbalist          (90
14 #> ...
```

Also emails, addresses, colors, coordinates, credit cards, etc.

Same idea in Python: `faker`

```
1 pip install Faker
```

```
1 from faker import Faker
2 fake = Faker()
3
4 fake.name()
5 #> 'Lucy Cechtelar'
6
7 fake.email()
8 #> 'tremblay.noe@example.org'
```

```
1 # reproducible
2 Faker.seed(4321)
3
4 # other locales
5 fake = Faker('fr_FR')
6
7 # build rows however you like
8 [fake.name() for _ in range(5)]
```

i `charlatan` is an R port of this package — same idea, either language
github.com/joke2k/faker

Your turn

15:00

Which airline should you avoid?

The code will run. The chart will look great. **The answer may be wrong.**

1. Ask Claude to rank the airlines by **mean departure delay** — joining `flights.csv` to `airlines.csv` for the names — and chart it
2. It will hand you a clean chart and a confident answer.
3. Then ask yourself three questions:
 - How many flights is each bar calculated from?
 - What happened to the **cancelled** flights?
 - Is any airline **missing** from the chart?
4. If something is wrong, tell Claude *specifically* what's wrong, and have it fix the chart.

Did the agent get any of this wrong?

Sample size

Alaska ranks **worst**, but only has **20 flights**.

Mesa: 10. Hawaiian: 11.

These are tiny samples

The vanished airline

SkyWest flew **once** but it was **cancelled**.

`mean()` returned `NaN`, so it dropped off the chart silently.

Cancellations

`na.rm = TRUE` deletes every cancelled flight.

Endeavor cancelled **7.3%** — none of those count as "late."

Every one of these is **invisible in the chart** and **invisible in the code**.

You only find them if you go looking.

Wrap-up

Today, in one slide

prompt → read the diff → verify → commit / push

- You can **make a repo and push** with a button
- You can **set up and direct Claude Code** — and read what it does
- You know `/init`, `/model`, `/effort`, `/clear`, `/compact`, `CLAUDE.md`, skills, and **modes**
- You know the agent **lies with confidence** — and you have a checklist

Before next week

- **HW2**: read forward on tidy data + joins, try the light practice, reflect
- Next week: **Cleaning, Reshaping & Joining** — same loop, on real data

Use [this link](#) to form teams
Review [potential project ideas](#)